b) Password Length Validation:

public int length() :

* It is a predefined non static method of String class, used to find out the length of the given String. The return type of this method is int.

Length.java

```
void main()
{
   String s1 = IO.readln("Enter some name :");
   int length = s1.length();
   IO.println("Length of "+s1+" is :"+length);
}
```

Password.java

```
void main()
{
     String password = IO.readln("Enter your password :");

     if(password.length()<8)
     {
          IO.println("Please provide strong password");
     }
}
```

c) Internet Connectivity Warning

Connectivity.java

```
void main()
{
     boolean isConnected = false;

     if(!isConnected)
     {
          IO.println("Internet Connection is not available");
     }
}
```

Assignments :

a) Speed Limit Alert (Vehicle System, Limit exceeded)
b) Bank Minimum Balance Alert
c) Invalid roll number

When to use if-else

The if-else statement is used when one action must be performed if the condition is true and a different action must be performed if the condition is false.

Syntax:

```
if(condition)
{
   // statements executed when condition is true
}
else
{
   // statements executed when condition is false
}
```

a) Number is even or odd
b) Eligible for Movie or not
c) Student Pass or Fail
d) Train berth available or not
e) Payment Status

When to use else if

If, we want to check **multiple conditions** one after another and wants to execute only one matching block among all the conditional blocks then we should use else-if.

Syntax:

```
if(condition1)
{
   // statements executed when condition1 is true
}
else if(condition2)
{
   // statements executed when condition2 is true
}
else if(condition3)
{
   // statements executed when condition3 is true
}
else
{
   // statements executed when all conditions are false
}
```

a) Student grading system [>=90 grade A, >=75 grade B, >=60 grade c else fail]

Grade.java

```
void main()
{
     int marks = Integer.parseInt(IO.readln("Enter your Marks :"));

     if(marks >=90)
     {
          IO.println("Grade A");
     }
     else if(marks >=75)
     {
          IO.println("Grade B");
     }
     else if(marks >=60)
     {
          IO.println("Grade C");
     }
     else
     {
      IO.println("Grade D");
     }
}
```

Traffic Signal System [String signal = "YELLOW";]

public boolean equals(Object obj)

* Predefined method of String class, Used to compare two String objects based on the **content, if two String object are exactly same (Case Sensitive) then it will return true otherwise false.**

Signal.java

```
void main()
{
     String signal = IO.readln("Color of the signal :").toUpperCase();

     if(signal.equals("YELLOW"))
     {
          IO.println("Please wait");
     }
     else if(signal.equals("RED"))
     {
          IO.println("Please Stop");
     }
     else if(signal.equals("GREEN"))
     {
          IO.println("Keep Moving");
     }
   else
     {
          IO.println("Invalid Color");
     }
}
```

Program to calculate telephone bill on the following criteria:
For 100 free call rentals = 360
For 101 - 250, 1 Rs per call
For 251 - unlimited, 1.2 Rs per call
Add 10% tax on total bill to calculate final bill

```
void main()
{

   int noOfCalls = Integer.parseInt(IO.readln("Enter your no of calls :"));
   double bill = 0.0;

    if(noOfCalls <=100)
      {
         bill = 360;
      }
      else if(noOfCalls <=250)
      {
      bill = 360 + (noOfCalls - 100) * 1.0;
      }
      else
      {
           bill = 360 + 150 + (noOfCalls - 250) * 1.2;
      }

    IO.println("Your bill is :"+bill);

      //Add 10 % extra tax and print the final bill

      double tax = bill * 0.10;

      IO.println("Your final bill is :"+(bill+tax));

}
```

Nested if:

If an **'if condition'** is placed inside another **if condition** then it is called Nested if.
In nested if condition, we have one outer if and one inner if condition, the inner if condition **will only execute** when outer if condition returns true.
We can also write if condition inside else block.

Syntax:

```
if(condition) //Outer if condition
{
   if(condition) //inner if condition
   {
   }
   else //inner else
   {
   }
}
else //outer else
{
   if(condition) //inner if condition
   {
   }
   else //inner else
   {
   }

}
```

Big.java

```
/*
Program to find out the biggest number among 3 numbers
*/
void main()
{
  int x = Integer.parseInt(IO.readln("Enter the value of x :"));
  int y = Integer.parseInt(IO.readln("Enter the value of y :"));
  int z = Integer.parseInt(IO.readln("Enter the value of z :"));

  int big = 0;

  if(x>y)
  {
    if(x>z)
      {
           big = x;
      }
      else
      {
      big = z;
      }
  }
  else  //already confirmed that y > x
  {
    if(y>z)
      {
           big = y;
      }
      else
      {
            big = z;
      }
  }

   IO.println("Biggest number is :"+big);

}
```

Note: In the above program to find out the biggest number among three number we need to take the help of **nested if condition** but the code becomes complex, to reduce the length of the code Logical Operator came into the picture.

Logical OR Conditional Operator:

It is used to combine or join the **multiple conditions** into a single statement.

It is also known as **short-Circuit** logical operator OR Conditional operator.

In Java we have 3 logical Operators

1) && (Logical AND Operator)

2) || (Logical OR Operator)

3) ? : Ternary (shorthand for if-then-else statement)

&&: All the conditions must be true. if the first expression is **false** it will not check right side expressions that is the reason It is short circuit operator

||: Among multiple conditions, at least one condition must be **true**. if the first expression is true it will not check right side expressions that is the reason It is short circuit operator

?: It consists of **three operands**. It is used to evaluate boolean expressions. The operator decides which value will be assigned to the variable. It is used to reduce the size **of if-else condition**.

Note: The && and || operator only works with boolean operand so the following code will not compile.

```
if(5 && 6)  //Invalid  becauae works with boolean value
{

}
```

And.java

```
void main()
{
   int a = 10, b = 20;

   boolean result = (a < b) && (b > 15);

   IO.println(result); //true


}
```

Big.java

```
void main()
{
  int a =Integer.parseInt(IO.readln("Enter the value of a :"));
  int b =Integer.parseInt(IO.readln("Enter the value of b :"));
  int c =Integer.parseInt(IO.readln("Enter the value of c :"));

    int big=0;

   if(a>b && a>c)
      {
         big = a;
      }
    else if(b > a && b > c)
      {
            big = b;
      }
      else
      {
     big = c;
    }
  IO.println("The big number is :"+big);
}
```

OR.java

```
void main()
{
  int x = 5, y = 3;

   boolean result = (x > 10) || (y < 5);

   IO.println(result);

}
```

Ternary Operator: [condition ? expression1 : expression2]

Ternary1.java

```
void main()
{
 int val1 = 100;
 int val2 = 200;
 boolean someCondition = false;

 //If consition is true expression1 will be assigned otherwise expression2
 int result = someCondition ? val1 : val2;
 IO.println(result);
}
```

Ternary2.java

```
//Max of two numbers

void main()
{
 int x = 300;
 int y = 200;

 int max = x>y ? x : y;

 IO.println(max);
}
```

Ternary3.java

```
//Max of three numbers

void main()
{
 int x = 300;
 int y = 700;
 int z = 900;

 int max = (x>y) ? ((x>z)?x:z) : ((y>z)?y:z);

 IO.println(max);
}
```